

Conditions for Test 1 Involvement

Truman Abbe

July 2026

DIY liquid rocket engines are a dangerous business, and essential safeguards are needed to ensure that serious injury, death, or severe legal consequences do not occur. If any one of the agreed conditions are broken, I, Truman Abbe, will walk away from the project. Any proposed amendments to this contract during the project will be considered soberly and agreed upon by all team members in writing. I, Truman Abbe, will offer my time and skill for the design, development, and first test of a small, bipropellant liquid rocket engine under the following conditions:

1. Test 1 will be executed through a local Tripoli chapter that will provide insurance, a test range, and expert consultation. All team members will register as Tripoli members. The project will be unaffiliated to Utah State University.
2. A safety officer role will not be designated to one team member, but all team members will equally and meticulously be safety aware throughout the duration of the project.
3. A soft deadline of August 31, 2026, will be set for the first test, and the project will be kept discrete.
4. The engine will have a maximum thrust of 50 N (≈ 10 lbf). The propellant tanks will be limited to 2.5 lbs for N₂O and 0.5 L for IPA. Considering a worst case scenario in which this amount of propellants are instantly mixed and ignited, preliminary calculations show that the energy output is equal to 3.9 lb of TNT with an energy yield of 7.34 MJ. Ear drum rupture would occur out to 9.7 m (32 ft). Fragmentation would have a range of 50 to 100 m. The fireball would have a radius of 2.1 m (7 ft).
5. I, Truman Abbe, will personally handle the propellants, design the plumbing system, and retain custody of the N₂O bottle and plumbing assembly.
6. Every action will be performed on paper before in real life. Work instructions will be written for every machining, assembly, and testing procedure. Critical opinion will be found through a CODR and CDR with field experts.
7. A team of a minimum of three engineers will undertake this endeavor, and all team members will unanimously authorize the test by their signatures, in addition to the signature of an advisor.
8. The test will follow FAA guidelines and BLM laws. Written permission from site administrators will be obtained. It may be required to purchase liability insurance for the test.
9. The engineers will brief their spouses on the systems and request their approval. It must be considered that an accident would not affect just the individual but also the family members reliant on that individual. Before traveling to the test site, the engineers will kneel and pray together. They will brief their systems to God and will ask His final approval. The test will not be conducted unless each engineer feels unequivocal divine approval.

Signature [Dax Davis]: _____ Date: _____

Signature [Truman Abbe]: _____ Date: _____

Witness: Printed: _____ Signature: _____ Date: _____